

Ramsha Khan

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Education

University of Mumbai, B.E in Computer Engineering

July 2023 – May 2026

- CGPA: 9.0/10.0

- **Coursework:** Machine Learning, Natural Language Processing, Big Data Analytics, Quantitative Analysis.

Technical Diploma in Computer Engineering

Aug 2020 – June 2023

- CGPA: 9.1/10.0

- **Coursework:** Foundations of Programming, Operating System, Computer Networks.

Experience

Data Analytics Intern, PrepInsta Technologies - Remote

Jan 2024 – Apr 2024

- Engineered data preprocessing pipeline for 8+ datasets using Python and SQL, reducing data inconsistencies by 90% and standardizing analysis-ready datasets.
- Performed in-depth exploratory data analysis to identify key trends and patterns that lead to actionable business insights.
- Recreated World Bank data visualization, following Sir Hans Rosling's style, showcasing data storytelling expertise.
- Developed an interactive Air Quality Analysis dashboard using Tableau, incorporating dynamic filters and diverse visualizations for comprehensive data storytelling. [View Dashboard](#)

Projects

FinetuneX

github FinetuneX

- Built FinetuneX from scratch, a modular framework for fine-tuning LLMs supporting **self-implemented multiple model architecture** with core transformer components includes **memory-efficient Grouped Query Attention** enabling faster inference with quality retention.
- Implemented **Supervised Fine-Tuning** post-training algorithms with optimized training techniques and extension for **Reinforcement Learning with Human Feedback** (DPO, PPO, RLOO) integration.
- Developed a user-facing platform that allows custom data set upload, training configuration, and base vs. fine-tuned model comparison.

FasalGuru

github FasalGuru

- Agricultural AI system built with the **Fine-tuned ResNet18** model for 22 crop disease classification and Ollama-hosted integrated LLMs for dynamic pest detection and prevention plans based on live weather data.
- Built ML models using Scikit-learn for irrigation and fertilizer prediction, enabling precision farming capabilities.
- Implemented **multilingual support** (200+ languages) for accessibility in diverse farming communities.

Achievements

- Led team to third place in MegaHack, a national-level hackathon at SJCEM, achieving a podium finish among 80+ teams.
- Secured first place among 100+ participants in the Technical Quiz, demonstrating strong problem solving and technical expertise.
- Earned Runner-Up in Code-a-thon, a competitive programming contest with 100+ participants, showcasing coding proficiency.
- Finalist in Technothon, an inter-college hackathon that demonstrates technical innovation and collaboration under tight deadlines.

Position of Responsibility

- As a Core Technical Member of Codecell at Rizvi College of Engineering, actively collaborated with the team to coordinate and improve technical events, workshops, and hackathons, fostering a technology-driven culture and driving innovation across the campus.

Technologies

Languages: Python, C++

Framework/Libraries: PyTorch, Tensorflow, Huggingface, scikit-learn, Docker

Technical Skills: Exploratory Data Analysis, Natural Language Processing, Machine Learning, Deep Learning, Reinforcement Learning, Generative AI (LLM, VLM)